

# Theorieexamen Advanced Programming

2021-2022, 1e zit

*You have three hours. Read the questions carefully and give a complete answer in no more than three paragraphs of two-three sentences per question.*

1. Indicate the main difference between the Lexical Analysis, Syntax Analysis and Context Analysis processes involved in the compilation of a program.
2. Provide the definition of "linkage". Describe the different types of linkage.
3. Describe the relationship between Storage Duration and the lifetime of a variable/object in a program.
4. Describe the use of the "mutable" qualifier in the context of physical and logic "constness".
5. Describe three reasons why it is crucial for the constructor to establish the invariant property of the constructed objects.
6. In the code below, indicate the operations or errors that may occur in the two lines following the definition of class S. Justify properly your answers:

```
struct S {  
    int a,b;  
}  
S obj1 ( 10, 20 );  
S obj2 { 10, 20 };
```

7. Describe the main difference regarding the internal operation of a copy constructor and a copy assignment.
8. Describe the difference between copy and move operations from the point of view of the arguments they receive as input.
9. Define the "hiding" concept that occurs in class inheritance. Explain with a code fragment how that hiding occurs.
10. Define the difference in terms of access control between the members of B and D between the following declarations.

```
class D : private B {...};  
class D : public B {...};
```

11. Describe the Liskov substitution principle.

12. Describe the "Factory" and "Abstract Factory" design patterns. Highlight the main difference between the two.
13. Indicate the main advantage of performing "concept checking".
14. Describe the main difference between the object oriented and the generic programming paradigms from the point of view of polymorphism.